

Effectivity

The following CT Equipment (see Table 1) is affected:

Product Name	Gantry Model #
Lightspeed Plus	2260619

Table 1

Purpose

The purpose of this FMI is to install hardware that will allow for "Field Balance of the Lightspeed Plus" CT Systems. This requires the removal of the existing "Trim Weights", installation of six threaded rods and performing the "Gantry Balance" procedure which uses "New Trim Weights" to balance the gantry.

Prerequisite FMIs

- FMI 25277 Lightspeed Plus 2.1 Software Upgrade

Related FMIs

- FMI 25277 Lightspeed Plus 2.1 Software Upgrade

Furnished Materials

FMI Kit part number 2306869 includes the following items:

Qty	Description	Part #
1	FMI Document	2305511-100
10	Lock Washer, M12	46-328432P7
10	Hex Nut, M12	46-328425P13
2	Threaded Rod (318 mm)	2276908-2
4	Threaded Rod (211 mm)	2276908-3
1	Loctite 271 (10 cc)	46-170648P2
1	Gantry Balance Weight Kit	2304948
1	3/8" drive, 19 mm (3/4 inch) crows foot	2309583
1	"Show Me" Video on CD-ROM	2318355-100

Tools Required

Qty	Description	Part #
1	Combination Wrench (19 mm)	Field Tool
1	Hex Key (10 mm)	Field Tool
1	Torque wrench kit	46-268445G1
1	3/8" 6 inch extension	Field Tool
1	3/8" drive, 19 mm deep well socket (6 point)	Field Tool
1	3/8" drive, 19 mm short well socket (6 point)	Field Tool
1	Tape Measure Metric	Field Tool

Manpower

Labor Estimates: 1 hour

Travel Estimates: 1 hour

General Information

WARNING: FAILURE TO PROPERLY TORQUE ROTATING GANTRY FASTNERS CAN RESULT IN SERIOUS PERSONAL INJURY AND EQUIPMENT DAMAGE.

Using Torque Wrench Extensions

This kit contains a $\frac{3}{4}$ inch (19 mm) crowfoot for $\frac{3}{8}$ inch socket drive. You cannot set your torque wrench directly. You must calculate the offset created by the extension to properly set the torque wrench dial. Failure to do this will result in excessive torque being applied to the threaded rods. Remember, torque is always measured at 90° angles. Deviation from this angle will reduce the applied torque regardless of the tool setting.

For the torque wrench specified in kit 46-268445G1, the Sturtevant Richmond model 3SDR-1200I, 200 – 1200 in-lb, GE part number 46-315668P2, the correct setting is:

62 N-m (45.7 ft-lb).

Since there are several versions of torque wrenches available, the following formula has been provided. This formula works for all "Click Type" or "Dial" Torque Wrenches. Refer to Figure 1.

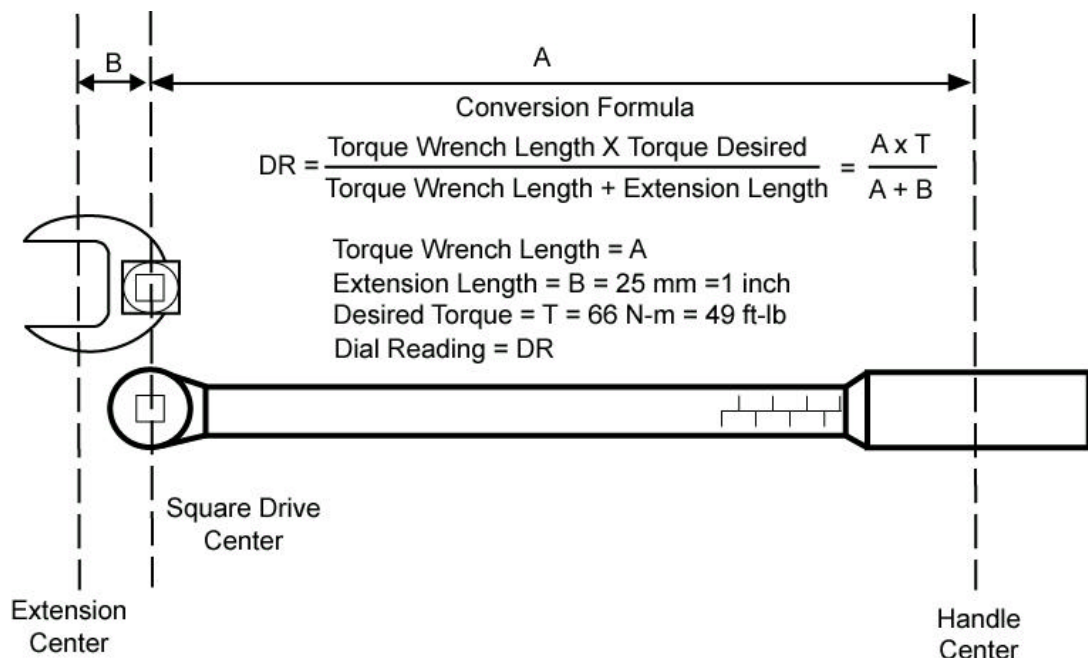


Figure 1

Procedure

Preparation

1. Remove gantry Front and Rear Covers.
2. Turn off all three STC Service switches (Axial Enable, HVDC Enable, 120 VAC).
3. Insure proper personal safety equipment is utilized  (eye protection and safety shoes).

Hardware Adjustment and Installation

Follow the steps listed below:

1. Remove the existing front and/or rear Balance Trim weights **ONLY** at the six (6) locations identified within this procedure. Refer to Figures 2 and 3. **DO NOT REMOVE WEIGHT FROM ANY OTHER LOCATIONS!**
 - Note: Original Trim weights can be round or rectangular. They can be found in locations other than those identified in this procedure. **DO NOT REMOVE WEIGHT FROM ANY OTHER LOCATIONS!**
 - Note: Save each grouping of weights including the mounting bolts and write down their locations. This will allow you to restore the gantry to its original "Balanced" condition if needed.
 - Note; you must have Lightspeed Plus Software version 2.1 to complete the balance portion of the FMI.
 - Note: The original weights and mounting hardware will not be used after the successful installation of this FMI and execution of the Gantry Balance Procedure available with the 2.1 software release. (*Helios_H2_02LESS_OC_Product 09/06/2001 LightSpeed Apps ver 14.13_2.8.2L_H2.1M4*)
2. Threaded rods must be installed at the following locations. Refer to Figures 2 and 3.
 - Note: Locations 60° and 308° are in forward production gantry's produced after August 1st, 2001. All gantries manufactured prior to this date will not accept threaded rods at these locations. The holes either do not exist or are not tapped completely through the casting. The need for these locations is a very low probability. These locations are identified in the event of a major component characteristic change in weight and center of gravity.
 - Note: The Gantry Balance software tool is designed for forward production systems. The potential exists that the calculated results will request weight to be added to locations 60° or 308°. In this event contact the Online Center for further instructions.

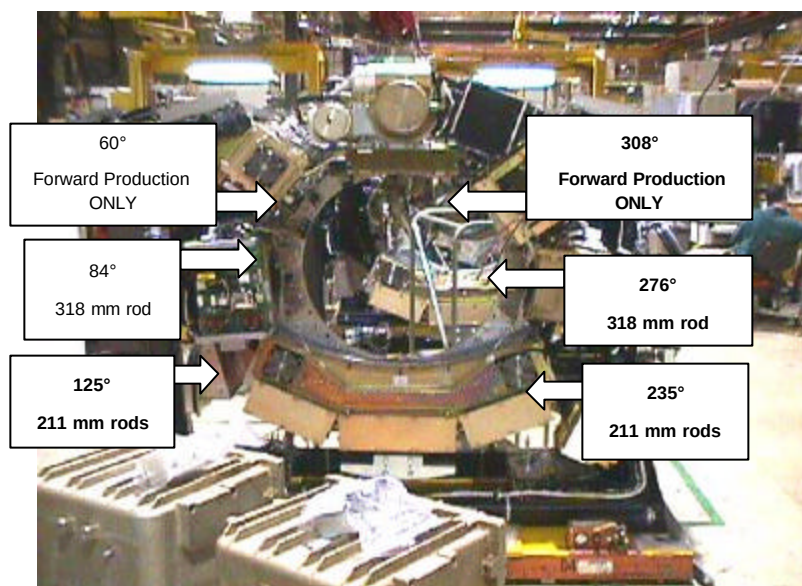


Figure 2

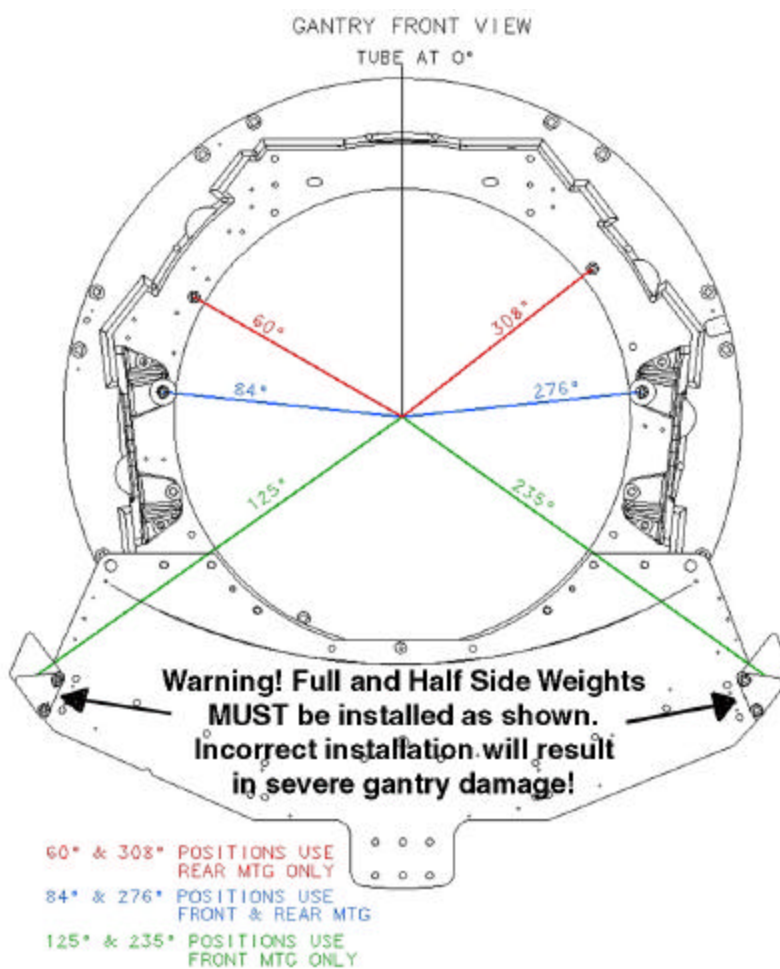


Figure 3

3. Start at the 84° and 276° locations. Refer to figures 4, 5, 6, 7, 8 and 9.

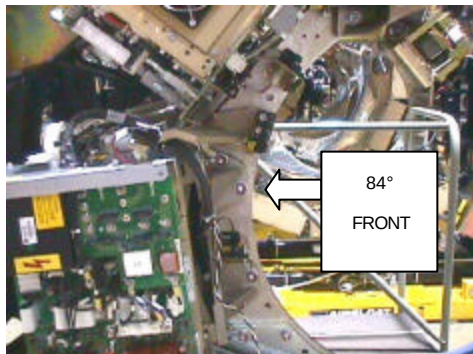


Figure 4

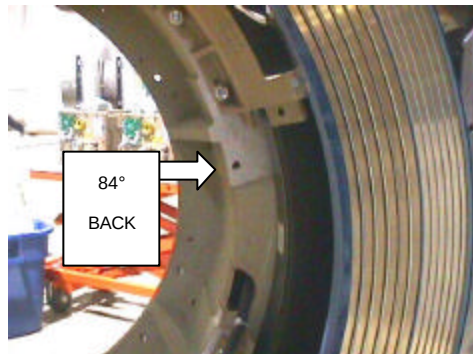


Figure 5

4. Remove the bolt and washer from the 276° location. Refer to Figure 8. For both 84° and 276° locations install one (1) 318 mm threaded rod (P/N 2276908-2) each as follows;
- 4.1. Place a pencil mark on each rod at 189 mm and 172 mm.
 - 4.2. Install the rod so that 189 +/- 2 mm extends out of the front of the rotational base.
 - Note: Using a tightened double nut at one rod end can help to thread the rods into these locations.
 - 4.3. Apply Loctite (P/N 46-170648P2) to the threads between the 189 mm and 172 mm pencil marks, 17 mm coverage.
 - 4.4. Now thread the rod such that 172 +/- 2 mm extends out of the front of the rotational base.
 - Note: Remove double nuts if used in step 4.1
 - 4.5. Install the Spacer (P/N 2302612) on the front of both rods.
 - 4.6. Install the wedge weight with pin (P/N 2302610) on the rear of both rods.
 - 4.7. Install a M12 lock washer (P/N 46-328432P7), and a M12 nut (P/N 46-328425P13) to the front and back of these rods.
 - 4.8. Torque to 66 N-m (49 ft-lb).
 - Note: Round Trim weights can be installed on the front and Wedge Trim weights can be installed on the back of these locations.
 - Note: This is the nut and washer storage location. These will be removed and used to secure the trim weights if need at these locations.

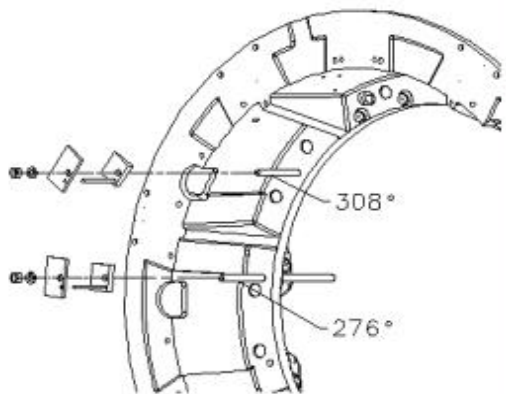


Figure 6

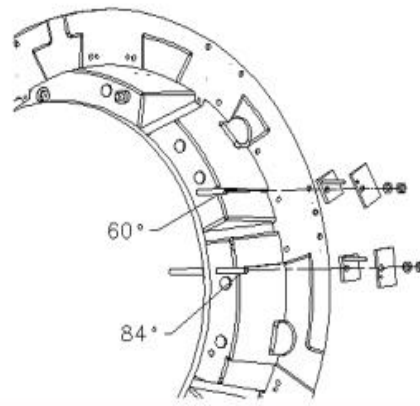


Figure 7

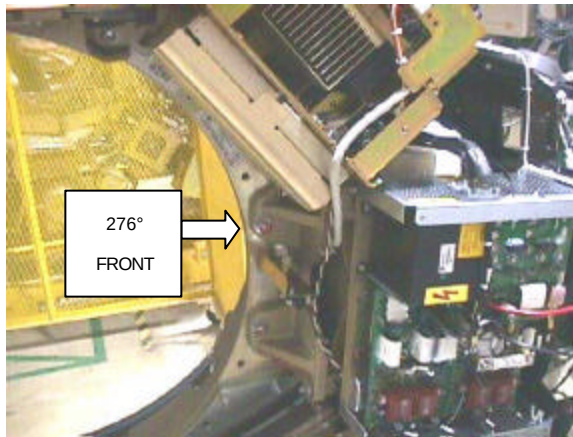


Figure 8

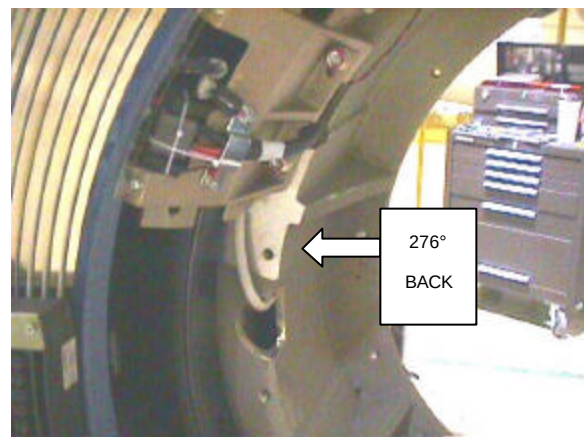


Figure 9

5. Next, will be the 125° and 235° locations. Refer to Figures 10, 11 and 12.

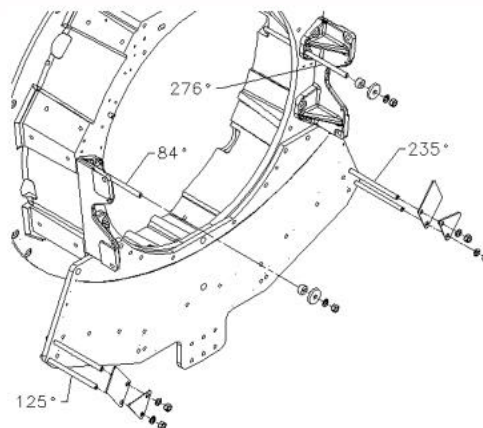


Figure 10

6. Install two 211 mm threaded rods (P/N 2276908-3) each at the 125° and 235° locations.
 - 6.1. Install the rods so that they are flush with the back of the detector plate.
 - 6.2. Apply Loctite (P/N 46-170648P2) to the threads that will engage the detector plate, about 10 mm.
 - 6.3. Now thread the rods so that 14 +/- 2 mm extend out of the back of the detector plate.
 - Note: This is to insure that there is full thread engagement of the lock washer and nut.
 - 6.4. Secure with one M12 nut (P/N 46-328425P13) and Loctite (P/N 46-170648P2) on the back of the detector plate.
 - Note: Do not install washer and nut on the front of the rods yet.
 - 6.5. Torque rear nuts to 66 N-m (49 ft-lb).
 - Note: Full Side and Half Side Trim weights can be installed only on the front of this location.

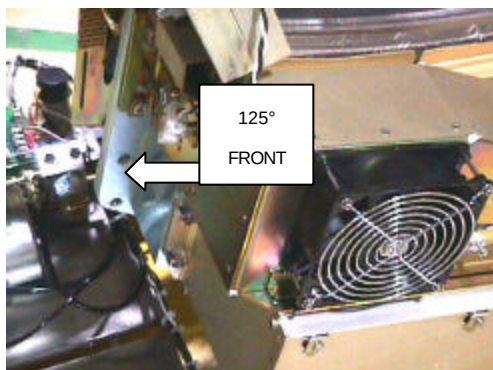


Figure 11

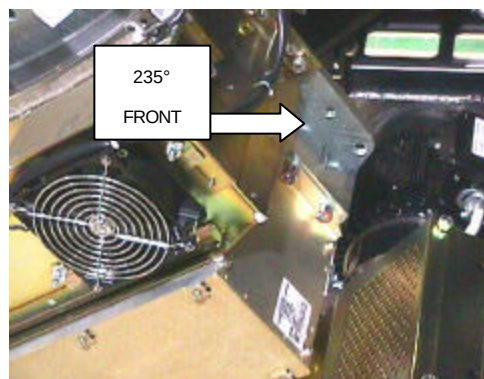


Figure 12

7. Install eight (8) Full Side Trim weights at the 235° location. Refer to Figure 3 for correct orientation of these weights.

WARNING: FAILURE TO INSTALL THESE WEIGHTS CORRECTLY WILL RESULT IN MAJOR DAMAGE TO THE RODS, DETECTOR PLATE AND STATIONARY GANTRY ASSEMBLIES.

- 7.1. Install one (1) M12 nut on a single rod; finger tight, to hold the trim weights in place.
 - This must be done so the Gantry Service Balance program will operate correctly.
 - Future balancing will not require this step as trim weights will already be installed.
 - Do not install the original balance weights. This will yield false results and waste time.
 - Rotate the Gantry by hand BEFORE proceeding with the software, to ensure that the threaded rods do not contact other components on the gantry.
8. Continue with the field balance software. Refer to Figure 13.
 - 8.1. On the right monitor click **Service**.
 - 8.2. On the Common Service Desktop click **Replacement Procedures**.
 - 8.3. Click on **Gantry Service Balance**.

9. Follow the program instructions display in the Status and Results frames.
Click on the **Help** button in the program to see the 4 steps needed to balance the gantry.
10. Discard the original weights and mounting hardware.
11. Verify all fastening hardware is tightened with the proper torque.
12. Place unused balance kit parts back into the original shipping box including the 19 mm (3/4 inch) Crowfoot wrench. Store onsite in the Service Cabinet or in the top of the PDU for future Gantry Balance needs.



Figure 13

FMI Completion

Fill out and return the FMI Completion Certification Report paperwork (or electronically close your FMI dispatch) in accordance with your local procedure.

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